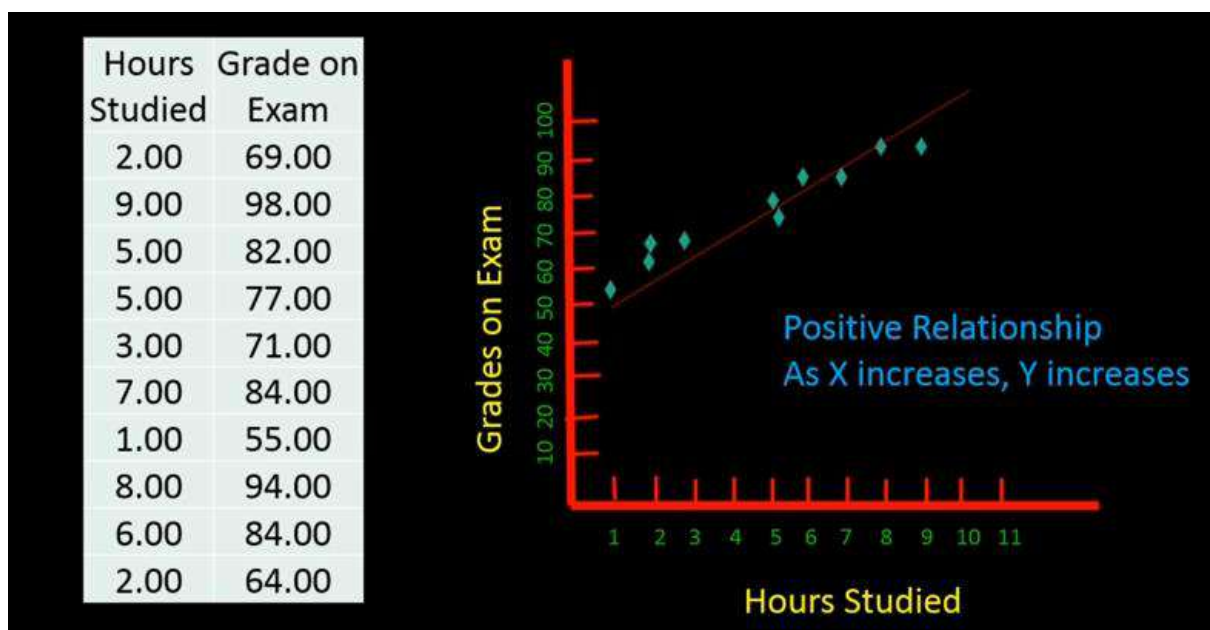


1. *What is Linear regression*
2. *Simple Linear regression*
3. *Multiple linear regression*
4. *Importance of Coeff and bias*
5. *We know Error, SSE, MSE, RMSE, Cost*
6. *Goal of Linear regression*
7. *How to find the Coeff using OLS*
8. *we know the Formuale b_0 and b_1*

OLS by hand



Hours(x)	Grade(y)
2	69
9	98
5	82
5	77
3	71
7	84
1	55
8	94
6	84
2	64

Here Input variable is : Hours (x)

Output variable is : Grade (y)

$$y = b_0 + b_1 * x_1$$

$$Grade = b_0 + b_1 * Hours$$

$$b_1 = \frac{\sum_i^n (\bar{y}_a - y_{ai}) * (\bar{x} - x_i)}{\sum_i^n (\bar{x} - x_i)^2}$$

$$b_0 = \bar{y}_a - b_1 * \bar{x}$$

what we need is:

$$\bar{x}$$

$$\bar{y}$$

$$y_i - \bar{y}$$

$$x_i - \bar{x}$$

$$(x_i - \bar{x}) * (y_i - \bar{y})$$

$$(\bar{x} - x_i)^2$$

Hours(X)	Grade(y)	$\bar{x} - x_i$	$(\bar{x} - x_i)^2$	$\bar{y} - y_i$	$(\bar{x} - x_i) * (\bar{y} - y_i)$
2	69	2.8	7.84	8.8	24.64
9	98	-4.2	17.64	-20.2	84.8
5	82	-0.2	0.04	-4.2	0.84
5	77	-0.2	0.04	0.8	-0.16
3	71	1.8	3.24	6.8	12.24

7	84	-2.2	4.84	-6.2	13.64
1	55	3.8	14.44	22.8	86.64
8	94	-3.2	10.24	-16.2	51.84
6	84	-1.2	1.44	-6.2	7.44
2	64	2.8	7.84	13.8	38.64
$\bar{x} = 4.8$	$\bar{y} = 77.8$		$\sum_i^n (\bar{x} - x_i)^2$ $= 67.24$		$\sum_i^n (\bar{y}_a - y_{ai}) * (\bar{x} - x_i)$ $= 320.6$

$$b_1 = \frac{\sum_i^n (\bar{y}_a - y_{ai}) * (\bar{x} - x_i)}{\sum_i^n (\bar{x} - x_i)^2} = \frac{320.6}{67.24} = 4.76$$

$$b_o = \bar{y}_a - b_1 * \bar{x} = 77.8 - 4.76 * 4.8 = 54.95$$

$$Grade = b_o + b_1 * Hours$$

$$Grade = 54.95 + 4.76 * Hours$$

$$GrE = 200 + 4.76 * Good\ ans$$

$$Grade = 54.95 + 4.76 * Hours$$

$$Hours = 0$$

$$Grade = 54.95 + 4.76 * 0 = 54.95\ garde$$

$$Hours = 1$$

$$Grade = 54.95 + 4.76 * 1 = 59grade$$

$$GrE = 200 + 4.76 * Good\ ans$$

